EE4047 High Voltage Techniques I - **Project Guideline**

Objective:

To introduce a High Voltage Test in use in the industry and research ,

and to introduce the capabilities and activities of a Large High Voltage Laboratory

Project Steps and Important Dates

Step 1) Join a group and select your HV test and lab (**until Friday, 12.05.23 – Week 10**)

Step 2) Make a group presentation to the class (**on Thursday, 25.05.23 – Week 12**)

PLEASE GO THE NEXT SLIDE

EE4047 High Voltage Techniques I - **Project Guideline**

Step 1 details:

Joining a group and selecting your topic (**until Friday, 12.05.23 – Week 10**)

- Form a group of 5 people,
- Select one of the experiments from the list given in the next slide,
- Find a large high voltage laboratory from internet, with enough information about its capabilities and activities .
- Send your group member details, selected experiment and HV lab to the course instructor (**alparslan.zehir@marmara.edu.tr**)

Note: Any students without a group or any group without a selected test or a lab by the deadline (12.05.23) will be automatically assigned with a group, test, lab by the course instructor.

PLEASE GO THE NEXT SLIDE

EE4047 High Voltage Techniques I - **Project Guideline**

The list of High Voltage tests (each group will pick a unique one)

- HV Test-1: Electrolytic tank experiment - mapping electric fields (**reserved**)
- HV Test-2: Measurement of high voltage with sphere gaps (**reserved**)
- HV Test-3: Impulse voltage tests (**reserved**)
- HV Test-4: Determination of breakdown voltage and breakdown strength in insulating oils (**reserved**)
- HV Test-5: Determination of voltage distribution and flashover voltage for insulator strings (**available**)
- HV Test-6: Measurement of dissipation factor and dielectric constant of solid dielectrics (**available**)

Note: The groups formed by students and their selected test and laboratories are listed in the document title ''PROJECT_Groups'' doument shared on the UES course page (under Week 0). The document is updated as new groups are formed and topics are selected by the students.

PLEASE GO THE NEXT SLIDE

EE4047 High Voltage Techniques I - **Project Guideline**

Step 2 details:

Make a group presentation to the class (**on Thursday, 25.05.23 – Week 12**)

The presentations will be max. 15 minutes long.

You can use any template.

Provide an informative presentation so that a person who knows nothing about your test and lab will be knowing the important information about it by the end of your presentation.

Each group members should take part in the presentation. You can decide with your group members about which part will be presented by which member.

Presentations will be in English.

If you need any further information please send a mail to: alparslan.zehir@marmara.edu.tr

EE4047 High Voltage Techniques I - **Project Guideline**

Step 2 details:

Some information you should provide about the HV test you selected:

- The purpose of the test, why it is needed
- The equipment used in the test and the test setup, use diagram-figure too for test setup
- Test steps, sequence of activities, what is done first, what is done after that and so on
- Formulas and calculations done as part of the test
- What are the results and findings of the test
- Give a standard about this test

PLEASE GO THE NEXT SLIDE

EE4047 High Voltage Techniques I - **Project Guideline**

Step 2 details:

Some information you should provide about the HV laboratory you selected:

- The description and history of the lab.(Lab. name, city, country, type (academic or industrial), when it was established, have it been renovated afterwards, is it still active and other) (include, up to 5 photos, showing the lab building and its interior, main spaces and equipment)
- What are the main testing equipment and their capabilities (max. V, max. current and other) in the laboratory?
- Which tests are being conducted in the lab?
- Which standards are followed during the tests?
- Are there any softwares or digital tools used in the lab.? What are they and for which activities they are used for?

If you need any further information please send a mail to: alparslan.zehir@marmara.edu.tr